

Table 1: Summary of Selected Literature

ID	Authors	Title	DOI
S1	Alanazi A.M.; Tuwajri I.; Khan F.; Alhamlan S.	Industrial Generative Artificial Intelligence Solutions at the Edge	10.2118/224681-MS
S2	Rubei R.; Di Salle A.; Bucaioni A.	LLM-Based Recommender Systems for Violation Resolutions in Continuous Architectural Conformance	10.1109/ICSA-C65153.2025.00063
S3	Zhang L.; Jia T.; Jia M.; Wu Y.; Liu H.; Li Y.	ScalaLog: Scalable Log-Based Failure Diagnosis Using LLM	10.1109/ICASSP49660.2025.10888670
S4	Das N.; Kotal A.; Roseberry D.; Joshi A.	Change Management using Generative Modeling on Digital Twins	10.1109/ISI58743.2023.10297181
S5	Oranekwu I.; Elluri L.; Batra G.	Automated Knowledge Framework for IoT Cybersecurity Compliance	10.1109/BigData62323.2024.10825755
S6	Berenguer A.; Morejón A.; Tomás D.; Mazón J.-N.	Using Large Language Models to Enhance the Reusability of Sensor Data	10.3390/s24020347
S7	Bento A.C.; De Los Santos-Resendiz C.; Segura-Prado L.E.; Garza-De La Rosa R.; De Jesus Hilario-Cruz R.; Gonzalez-Vazquez S.	Experimental Results with the Use of OpenAI with IoT Classes for Water Analysis in Rainfall	10.1109/icSmartGrid61824.2024.10578189
S8	Tafferner Z.; Illés B.; Krammer O.; Géczy A.	Can ChatGPT Help in Electronics Research and Development? A Case Study with Applied Sensors	10.3390/s23104879
S9	Fathy Y.; Jaber M.; Brintrup A.	Learning with Imbalanced Data in Smart Manufacturing: A Comparative Analysis	10.1109/ACCESS.2020.3047838
S10	Zosimadis I.; Stamelos I.	LLM-Enhanced Test Case Prioritization for Complex Software Systems	10.1145/3716554.3716561
S11	Peer J.; Mordecai Y.; Reich Y.	NLP4ReF: Requirements Classification and Forecasting: From Model-Based Design to Large Language Models	10.1109/AERO58975.2024.10521022
S12	Yang H.; Zhou Y.; Liang T.; Kuang L.	ChatDL: An LLM-Based Defect Localization Approach for Software in IIoT Flexible Manufacturing	10.1109/JIOT.2025.3531512
S13	Sabatucci L.; Cossentino M.; Di Napoli C.; Susi A.	Enhancing Device-Goal-Norm Modeling for Ambient Assisted Living with Large Language Models	10.1145/3672608.3707813
S14	Alanazi A.M.; Altuwajri I.A.; Bagabas A.M.; Khan F.; Otaibi B.	Generative Edge Artificial Intelligence at Oil Rigs	10.2118/225701-MS
S15	Yang H.; Siew M.; Joe-Wong C.	An LLM-Based Digital Twin for Optimizing Human-in-the Loop Systems	10.1109/FMSys62467.2024.00009
S16	Aramkul S.; Sugunnasil P.	Intelligent IoT framework with GAN-synthesized images for enhanced defect detection in manufacturing	10.1111/coin.12619
S17	Maciej Hojda	Using Large Language Models for Processing Sensor Data	10.3390/s25144380
S18	Alberto Berenguer, Adriana Morejón, David Tomás and Jose-Norberto Mazón	Leveraging Large Language Models for Sensor Data Retrieval	10.3390/app14062506
S19	Yauri, Ricardo; Espino, Rafael	Generative Language Model Technology Integrated into an IoT Device for the Development of a Voice Assistant	10.37394/23202.2024.23.54

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ID	Authors	Title	DOI
S20	Shekhar S.; McCafferty W.	An LLM-based Semantic Middleware for Energy Asset as a Service (EAaaS)	10.1145/3774900.3776633
S21	Zhang Z.; Du X.; Su W.; Camburn B.	Digital Designer: Emulation of Empathy-Driven Design with Generative AI	10.1109/LCIoT64881.2025.11118557
S22	Marathe A.; Vaniya P.; Motale V.; Thokal P.; Mehare Y.; Wayal S.	IoT-Enabled Real-Time Milk Quality Assessment System using Multi-Sensor Fusion and XAI for Microbial Load Prediction	10.1109/ICESC65114.2025.11212238
S23	Chandrakumaran A.	Enhancing SAP Predictive Maintenance and Industrial Analytics with LLM-Driven Automation	10.1109/MPCON66082.2025.11256577
S24	Ikegami Y.; Hasegawa K.; Hidano S.; Fukushima K.; Hashimoto K.; Togawa N.	Automated Security Compliance Evaluation Using Hierarchical RAG for IoT Devices with Large-Scale Documentation	10.1109/IOTSMS68530.2025.11408530
S25	Wang N.	Lightweight Vision-Language Model for Fashion Design in IoT Environment	10.1002/itl2.70140
S26	Tikka P.; Karjalainen J.; Alesani A.; Goriachev V.	Large Language Model Hallucination Mitigation in Three Industrial Use Cases	10.1109/ACCESS.2026.3659997
S27	Zaitsev A.	Resilience-oriented lifecycle automation of structural RC elements via a multi-agent system framework	10.26599/JIC.2026.9180111
S28	Jabrw S.K.; Sarhan Q.I.	Evaluating Large Language Models for Arduino Code Generation	10.14500/aro.12344
S29	Xia Z.; Zhong B.; Zhang S.; Zhao T.; Skibniewski M.J.	Graph-driven embedding reinforcement and traceable LLM agent for reliable element alignment in construction report generation	10.1016/j.autcon.2026.106816
S30	Ramalingam V.; Kumar B.; Gupta S.K.; Alsekait D.M.; AbdElminaam D.S.	A hybrid federated learning framework with generative AI for privacy-preserving and sustainable security in IOT-enabled smart environments	10.1038/s41598-025-31769-6
S31	Zhang Z.; Li Z.; Xu S.; Zhang F.; Song Z.; Fang D.; Chen Y.; Sun L.	NetID-GPT: Adapting large language models for large-scale internet-connected device identification	10.1016/j.comnet.2026.112149
S32	Babiuch M.; Smutný P.	Benchmarking Large Language Models for Embedded Systems Programming in Microcontroller-Driven IoT Applications	10.3390/fi18020094
S33	Peng Y.; Gong K.; Gao Y.; Wang H.; Dong W.	IoTMigrator: LLM-driven Embedded IoT Code Migration across Different OSES for Cloud-device Integration	10.18653/v1/2025.findings-emnlp.1048
S34	Masoumzadeh S.	Detecting Vulnerabilities from Issue Reports for Internet-of-Things	10.1109/ASE63991.2025.00394
S35	Bonanno M.; Russo M.; Santoro C.; Santoro F.F.; Tudisco A.	BDI-Driven Indoor Positioning and Assistance via Hermes Mesh Networks and LLM Interfaces	10.1109/PICOM68402.2025.00056
S36	Myakala P.K.; Kamatala S.; Naayini P.	Edge AI and Federated LLMOps for Latency-Critical IoT Systems in Smart Infrastructure	10.1109/CONIT65521.2025.11167465

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ID	Authors	Title	DOI
S37	Nguno V.; Gertze R.; Karamata R.; Mukamana A.; Kamukuju F.; Till S.; Indongo N.; Kandjengo S.; Rodil K.; Smith R.C.; Winschiers-Theophilus H.	Co-Designing a Web-Based Interface for an AI-IoT Hydroponics System with an Indigenous Community in Namibia	10.1109/ETNCC66224.2025.11299703
S38	Peer J.; Mordecai Y.; Reich Y.	Self-Generative Requirements Engineering	10.1109/SMC58881.2025.11343115
S39	Sebastien Gerin, Laurent Joblot, Nisrine Makhoul, Frederic Merienne	Ontology-driven LLM framework for knowledge graph in smart buildings	10.22260/ISARC2025/0137
S40	Zhenhao Tian , Yi He , Nuo Zhang , Qixiao Lin , Hetian Shi , Jianwei Zhuge , Jian Mao , Member, IEEE, and Deliang Chang	BLMProbe: Enhancing Internet-Connected Device Discovery by Automated Device Labeling and Label Migration	10.1109/TIFS.2025.3587211
S41	L. Coppolino, A. Iannaccone, R. Nardone, A. Petruolo	Asset Discovery in Critical Infrastructures: An LLM-Based Approach	10.3390/electronics14163267
S42	Y. Ikegami, K. Hasegawa, S. Hidano, K. Fukushima, K. Hashimoto, and N. Togawa	Security Conformance Scoring for IoT Devices Through Documentation Analysis Using Large Language Models	10.1109/ICCE-Berlin67488.2025.11277466